CerebroTwin -- Physiological Digital Twin Architecture

Non-Cerebral Cardiovascular & Respiratory Biosignals --> 0-60s Orthostatic MCAv Instability Forecasting

Static Baseline Covariates Route (Non-Overlapping Orthogonal Flow)

1A. Clinical Inputs
Age | Sex | T2D Status
HbA1c | Hypertension
Medications (Antihypertensives)
Resting MAP & Heart Rate

1B. Continuous Dynamic Signals
Beat-to-Beat Arterial BP (MAP)
ECG / Heart Rate Dynamics
Respiration Rate | EtCO2 Waveform
0-5s Post-Stand Orthostatic Drop

2. Preprocessing & Alignment
Beat-to-Beat Synchronization
Dynamic Window Alignment (0-60s)
Signal QC & Outlier Imputation
Z-score / Robust Feature Scaling

3. CerebroTwin State Core (27 Features)
Baroreflex Sensitivity Proxy (BRS)
Cerebral Autoregulation Index (CAI)
Delta HR/dt | Delta MAP/dt Slopes
EtCO2 Vasoreactivity Gradients
Autonomic Tone & Heart Rate Variability

4. Multi-Target ML Forecasting Engine
Ensemble of Progression Baselines & Production Quantile Gradient Boosting

Model 0: Mean
Heuristic Control Baseline
Fixed Group Average

Model 1: Ridge
Linear Regularization
L2 Penalty Shrinkage

Model 2: Quantile
Multi-Output XGBoost
90% Prediction Intervals

5. 0-60s MCAv Forecast Outputs
(19 Orthostatic Targets + 90% Prediction Intervals)

Predicted Peak Relative Drop (pct_mca_drop)
Minimum Velocity / Nadir (MCAv_min cm/s) & Time-to-Nadir
Cumulative Cerebral Hypoperfusion Deficit (AUC < Baseline)
Trajectory Recovery Status: Rapid vs Delayed Compensator
Clinical Alert: Normal vs High-Deviation Postural Drop

GroupKFold Benchmark

6. Ground Truth Validation Gate
(TCD MCAv Strictly Isolated from Runtime Inputs)

Measured Transcranial Doppler (TCD) Ultrasound
Zero-Leakage 5-Fold GroupKFold Subject Split
Subject Exists in Train OR Test -- Never Both
Primary Peak R2 = 0.735 | Drop MAE = 4.816%
Continuous 0-60s Trajectory RMSE = 4.475%

7. Clinician Dashboard & Counterfactual What-If Simulator
(Live Deployed on Streamlit Community Cloud)

Interactive Patient Waveform Replay | Real-Time 0-60s MCAv Curve Overlay & 90% Confidence Ribbons
Counterfactual Simulation: 'What if HbA1c drops to 7.0%?' | 'What if Antihypertensive Dosage Adjusted?'
SHAP Feature Importance Explanations (TreeExplainer Waterfall) | Orthostatic Syncope Deficit Risk Scoring
Live Cloud App: https://cerebrotwin-cxdos3it3cqlvhu6eftmtt.streamlit.app/

Architecture Legend

- 1A/1B. Clinical & Signals
- 2. Preprocessing & Sync
- 3. Digital Twin State
- 4. Multi-Target ML
- 5. Forecast Outputs
- 6. Ground Truth Isolation
- 7. Clinician Dashboard

Validation Performance
5-Fold Subject GroupKFold

Primary Peak R2: 0.735
Drop MAE: 4.816%
Trajectory RMSE: 4.475%
Uncertainty: 90% Quantile CI

PhysioNet Dataset
Cerebral Perfusion & Vasoregulation in Type 2 Diabetes (v1.0.1)

Primary Repository URL:
physionet.org/content/cerebral-perfusion-diabetes/1.0.1/

Subject Cohort: T2D & Controls
Continuous TCD, ABP, ECG, EtCO2
Zero Pre-Stand Runtime Leakage

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